JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Application of Derivatives

JEE Main Mathematics · Previous Year Questions with Answer Key

229	144	2002 – 2026	60	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer)	142 questions
Numerical	27 questions
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Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 142 questions

Q1. JEE Main 2026 (Online) 5th April Morning Shift

Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function such that $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}$ for all $x,y\in\mathbb{R}$, and $f'(0)=3$. Then the minimum value of the function $g(x)=3+e^x f(x)$, is:

(A) $3\left(\frac{e+1}{e}\right)$ (B) $3\left(\frac{e-1}{e}\right)$ (C) $\frac{3-e}{e}$ (D) 3e

Q2. JEE Main 2026 (Online) 2nd April Evening Shift

Let $f(x)$ be a polynomial of degree 5, and have extrema at $x=1$ and $x=-1$. If $\lim\limits_{x\to 0}\left(\frac{f(x)}{x^3}\right)=-5$, then $f(2)-f(-2)$ is equal to:

(A) 0 (B) 50 (C) 92 (D) 112

Q3. JEE Main 2026 (Online) 2nd April Morning Shift

The number of critical points of the function

$f(x)=\begin{cases}\left|\frac{\sin x}{x}\right|, & x\neq 0\\ 1, & x=0\end{cases}$ in the interval $(-2\pi,2\pi)$ is equal to :

(A) 1 (B) 3 (C) 5 (D) 7

Q4. JEE Main 2026 (Online) 24th January Evening Shift

Consider the following three statements for the function $f:(0,\infty)\to\mathbb{R}$ defined by $f(x)=|\log_e x|-|x-1|$:

(I) f is differentiable at all $x>0$.

(II) f is increasing in $(0,1)$.

(III) f is decreasing in $(1,\infty)$.

Then.

(A) Only (I) is TRUE. (B) Only (I) and (III) are TRUE.

(C) Only (II) and (III) are TRUE. (D) All (I), (II) and (III) are TRUE.

Q5. JEE Main 2026 (Online) 23rd January Evening Shift

The least value of $\left(\cos^2\theta-6\sin\theta\cos\theta+3\sin^2\theta+2\right)$ is

(A) $4-\sqrt{10}$ (B) -1 (C) $4+\sqrt{10}$ (D) 1

Q6. JEE Main 2026 (Online) 23rd January Morning Shift

Let α and β respectively be the maximum and the minimum values of the function $f(\theta) = 4\left(\sin^4\left(\frac{7\pi}{2} - \theta\right) + \sin^4(11\pi + \theta)\right) - 2\left(\sin^6\left(\frac{3\pi}{2} - \theta\right) + \sin^6(9\pi - \theta)\right), \theta \in \mathbf{R}$.

Then $\alpha + 2\beta$ is equal to :

(A) 6 (B) 5 (C) 4 (D) 3

Q7. JEE Main 2026 (Online) 22nd January Morning Shift

Let $f(x) = x^{2025} - x^{2000}, x \in [0, 1]$ and the minimum value of the function $f(x)$ in the interval $[0, 1]$ be $(80)^{80}(n)^{-81}$. Then n is equal to

(A) -40 (B) -41 (C) -80 (D) -81

Q8. JEE Main 2026 (Online) 21st January Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$ and $f'(a - 1) = 0$, where a is a real number.

Let $g(x) = f(\tan^2 x - 2\tan x + a),\ 0 < x < \frac{\pi}{2}$.

Consider the following two statements:

(I) g is increasing in $\left(0, \frac{\pi}{4}\right)$

(II) g is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Then,

(A) Both (I) and (II) are True (B) Neither (I) nor (II) is True

(C) Only (I) is True (D) Only (II) is True

Q9. JEE Main 2025 (Online) 8th April Evening Shift

Let the function $f(x) = \frac{x}{3} + \frac{3}{x} + 3, x \neq 0$ be strictly increasing in $(-\infty, \alpha_1) \cup (\alpha_2, \infty)$ and strictly decreasing in $(\alpha_3, \alpha_4) \cup (\alpha_4, \alpha_5)$. Then $\sum_{i=1}^{5} \alpha_i^2$ is equal to

(A) 48 (B) 40 (C) 36 (D) 28

Q10. JEE Main 2025 (Online) 7th April Evening Shift

Let **f : $\mathbb{R} \to \mathbb{R}$** be a polynomial function of degree four having extreme values at **x = 4** and **x = 5**. If $\lim_{x \to 0} \frac{f(x)}{x^2} = 5$, then **f(2)** is equal to :

(A) 8 (B) 10 (C) 12 (D) 14

Q11. JEE Main 2025 (Online) 7th April Morning Shift

Let $x=-1$ and $x=2$ be the critical points of the function $f(x)=x^3+ax^2+b\log_e|x|+1, x\neq 0$. Let m and M respectively be the absolute minimum and the absolute maximum values of f in the interval $\left[-2,-\frac{1}{2}\right]$. Then $|\mathrm{M}+m|$ is equal to (Take $\log_e 2=0.7$) :

(A) 21.1 (B) 19.8 (C) 22.1 (D) 20.9

Q12. JEE Main 2025 (Online) 4th April Evening Shift

Let $\mathrm{a}>0$. If the function $f(x)=6x^3-45\mathrm{a}x^2+108\mathrm{a}^2x+1$ attains its local maximum and minimum values at the points x_1 and x_2 respectively such that $x_1x_2=54$, then $\mathrm{a}+x_1+x_2$ is equal to :

(A) 15 (B) 13 (C) 24 (D) 18

Q13. JEE Main 2025 (Online) 3rd April Evening Shift

The shortest distance between the curves $y^2=8x$ and $x^2+y^2+12y+35=0$ is:

(A) $2\sqrt{3}-1$ (B) $2\sqrt{2}-1$ (C) $3\sqrt{2}-1$ (D) $\sqrt{2}$

Q14. JEE Main 2025 (Online) 3rd April Evening Shift

Let $f:\mathrm{R}\to\mathrm{R}$ be a function defined by $f(x)=\big||x+2|-2|x|\big|$. If m is the number of points of local minima and n is the number of points of local maxima of f, then $m+n$ is

(A) 3 (B) 4 (C) 2 (D) 5

Q15. JEE Main 2025 (Online) 2nd April Morning Shift

If the function $f(x)=2x^3-9ax^2+12\mathrm{a}^2x+1$, where $\mathrm{a}>0$, attains its local maximum and local minimum values at p and q , respectively, such that $\mathrm{p}^2=\mathrm{q}$, then $f(3)$ is equal to :

(A) 55 (B) 37 (C) 10 (D) 23

Q16. JEE Main 2025 (Online) 28th January Morning Shift

The sum of all local minimum values of the function

$$\mathrm{f}(x)=\begin{cases}1-2x, & x<-1\\ \frac{1}{3}(7+2|x|), & -1\le x\le 2\\ \frac{11}{18}(x-4)(x-5), & x>2\end{cases}$$

is

(A) $\frac{167}{72}$ (B) $\frac{157}{72}$ (C) $\frac{171}{72}$ (D) $\frac{131}{72}$

Q17. JEE Main 2025 (Online) 24th January Evening Shift

Let $(2,3)$ be the largest open interval in which the function $f(x)=2\log_e(x-2)-x^2+ax+1$ is strictly increasing and (b, c) be the largest open interval, in which the function $g(x)=(x-1)^3(x+2-a)^2$ is strictly decreasing. Then $100(a+b-c)$ is equal to :

(A) 360 (B) 420 (C) 160 (D) 280

Q18. JEE Main 2025 (Online) 24th January Morning Shift

Consider the region $R=\left\{(x,y): x\leq y\leq 9-\frac{11}{3}x^2, x\geq 0\right\}$. The area of the largest rectangle of sides parallel to the coordinate axes and inscribed in R , is:

(A) $\frac{821}{123}$ (B) $\frac{567}{121}$ (C) $\frac{730}{119}$ (D) $\frac{625}{111}$

Q19. JEE Main 2025 (Online) 23rd January Evening Shift

A spherical chocolate ball has a layer of ice-cream of uniform thickness around it. When the thickness of the ice-cream layer is 1 cm , the ice-cream melts at the rate of $81\ \mathrm{cm}^3/\mathrm{min}$ and the thickness of the ice-cream layer decreases at the rate of $\frac{1}{4\pi}\ \mathrm{cm/min}$. The surface area (in cm^2) of the chocolate ball (without the ice-cream layer) is :

(A) 128π (B) 196π (C) 225π (D) 256π

Q20. JEE Main 2025 (Online) 22nd January Evening Shift

Let $f(x)=\int_0^{x^2}\frac{t^2-8t+15}{e^t}dt, x\in\mathbf{R}$. Then the numbers of local maximum and local minimum points of f, respectively, are :

(A) 3 and 2 (B) 2 and 2 (C) 2 and 3 (D) 1 and 3

Q21. JEE Main 2024 (Online) 8th April Evening Shift

If the function $f(x)=2x^3-9ax^2+12a^2x+1, a>0$ has a local maximum at $x=\alpha$ and a local minimum at $x=\alpha^2$, then α and α^2 are the roots of the equation :

(A) $x^2-6x+8=0$ (B) $8x^2-6x+1=0$ (C) $8x^2+6x-1=0$ (D) $x^2+6x+8=0$

Q22. JEE Main 2024 (Online) 8th April Morning Shift

Let $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$. The number of points of local maxima of f in interval $(0,2\pi)$ is

(A) 1 (B) 3 (C) 4 (D) 2

Q23. JEE Main 2024 (Online) 8th April Morning Shift

The number of critical points of the function $f(x)=(x-2)^{2/3}(2x+1)$ is

(A) 2 (B) 1 (C) 0 (D) 3

Q24. JEE Main 2024 (Online) 8th April Morning Shift

For the function $f(x) = (\cos x) - x + 1, x \in \mathbb{R}$, between the following two statements

(S1) $f(x) = 0$ for only one value of x in $[0, \pi]$.

(S2) $f(x)$ is decreasing in $\left[0, \frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2}, \pi\right]$.

(A) Both (S1) and (S2) are incorrect.

(B) Only (S1) is correct.

(C) Only (S2) is correct.

(D) Both (S1) and (S2) are correct.

Q25. JEE Main 2024 (Online) 6th April Evening Shift

If the function $f(x) = \left(\frac{1}{x}\right)^{2x}; x > 0$ attains the maximum value at $x = \frac{1}{\mathrm{e}}$ then :

(A) $\mathrm{e}^{\pi} < \pi^{\mathrm{e}}$

(B) $\mathrm{e}^{2\pi} < (2\pi)^{\mathrm{e}}$

(C) $(2e)^{\pi} > \pi^{(2e)}$

(D) $\mathrm{e}^{\pi} > \pi^{\mathrm{e}}$

Q26. JEE Main 2024 (Online) 6th April Morning Shift

The interval in which the function $f(x) = x^x, x > 0$, is strictly increasing is

(A) $(0, \infty)$

(B) $\left(0, \frac{1}{e}\right]$

(C) $\left[\frac{1}{e^2}, 1\right)$

(D) $\left[\frac{1}{e}, \infty\right)$

Q27. JEE Main 2024 (Online) 5th April Morning Shift

Let a rectangle ABCD of sides 2 and 4 be inscribed in another rectangle PQRS such that the vertices of the rectangle ABCD lie on the sides of the rectangle PQRS. Let a and b be the sides of the rectangle PQRS when its area is maximum. Then $(a+b)^2$ is equal to :

(A) 64

(B) 80

(C) 60

(D) 72

Q28. JEE Main 2024 (Online) 5th April Morning Shift

Let $f(x) = x^5 + 2x^3 + 3x + 1, x \in \mathbf{R}$, and $g(x)$ be a function such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. Then $\frac{g(7)}{g'(7)}$ is equal to :

(A) 42

(B) 7

(C) 1

(D) 14

Q29. JEE Main 2024 (Online) 5th April Morning Shift

For the function

$f(x) = \sin x + 3x - \frac{2}{\pi}\left(x^2 + x\right)$, where $x \in \left[0, \frac{\pi}{2}\right]$,

consider the following two statements :

(I) f is increasing in $\left(0, \frac{\pi}{2}\right)$.

(II) f' is decreasing in $\left(0, \frac{\pi}{2}\right)$.

Between the above two statements,

(A) only (I) is true.

(B) both (I) and (II) are true.

(C) only (II) is true.

(D) neither (I) nor (II) is true.

Q30. JEE Main 2024 (Online) 4th April Evening Shift

Let $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f, then $\alpha^2+2\beta^2$ is equal to

(A) 42 (B) 38 (C) 24 (D) 44

Q31. JEE Main 2024 (Online) 4th April Morning Shift

Let the sum of the maximum and the minimum values of the function $f(x)=\frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$, where $\gcd(\mathrm{m},\mathrm{n})=1$. Then $\mathrm{m}+\mathrm{n}$ is equal to :

(A) 217 (B) 182 (C) 201 (D) 195

Q32. JEE Main 2024 (Online) 1st February Morning Shift

If $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2, \forall x\neq 0$ and $y=9x^2f(x)$, then y is strictly increasing in :

(A) $\left(0,\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$ (B) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

(C) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$ (D) $\left(-\infty,\frac{1}{\sqrt{5}}\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$

Q33. JEE Main 2024 (Online) 31st January Evening Shift

If the function $f:(-\infty,-1]\to(a,b]$ defined by $f(x)=e^{x^3-3x+1}$ is one - one and onto, then the distance of the point $P(2b+4,a+2)$ from the line $x+e^{-3}y=4$ is :

(A) $2\sqrt{1+e^6}$ (B) $\sqrt{1+e^6}$ (C) $3\sqrt{1+e^6}$ (D) $4\sqrt{1+e^6}$

Q34. JEE Main 2024 (Online) 30th January Evening Shift

Let $f(x)=(x+3)^2(x-2)^3, x\in[-4,4]$. If M and m are the maximum and minimum values of f, respectively in $[-4,4]$, then the value of $M-m$ is

(A) 108 (B) 392 (C) 608 (D) 600

Q35. JEE Main 2024 (Online) 30th January Morning Shift

The maximum area of a triangle whose one vertex is at $(0,0)$ and the other two vertices lie on the curve $y=-2x^2+54$ at points (x,y) and $(-x,y)$, where $y>0$, is :

(A) 108 (B) 122 (C) 88 (D) 92

Q36. JEE Main 2024 (Online) 29th January Evening Shift

The function $f(x)=\frac{x}{x^2-6x-16}, x\in\mathbb{R}-\{-2,8\}$

(A) decreases in $(-\infty,-2)\cup(-2,8)\cup(8,\infty)$

(B) increases in $(-\infty,-2)\cup(-2,8)\cup(8,\infty)$

(C) decreases in $(-2,8)$ and increases in $(-\infty,-2)\cup(8,\infty)$

(D) decreases in $(-\infty,-2)$ and increases in $(8,\infty)$

Q37. JEE Main 2024 (Online) 29th January Evening Shift

The function $f(x) = 2x + 3(x)^{\frac{2}{3}}, x \in \mathbb{R}$, has

(A) exactly one point of local minima and no point of local maxima

(B) exactly one point of local maxima and exactly one point of local minima

(C) exactly two points of local maxima and exactly one point of local minima

(D) exactly one point of local maxima and no point of local minima

Q38. JEE Main 2024 (Online) 29th January Morning Shift

Consider the function $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ defined by $f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$. Consider the statements

(I) The curve $y = f(x)$ intersects the x-axis exactly at one point.

(II) The curve $y = f(x)$ intersects the x-axis at $x = \cos\frac{\pi}{12}$.

Then

(A) Both (I) and (II) are correct.

(B) Only (I) is correct.

(C) Both (I) and (II) are incorrect.

(D) Only (II) is correct.

Q39. JEE Main 2024 (Online) 27th January Evening Shift

Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3 - x)$ and $f''(x) > 0$ for all $x \in (0, 3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is :

(A) 0 (B) 24 (C) 18 (D) 20

Q40. JEE Main 2023 (Online) 13th April Morning Shift

$\max\limits_{0 \le x \le \pi} \left\{x - 2\sin x \cos x + \frac{1}{3}\sin 3x\right\} =$

(A) $\frac{5\pi+2+3\sqrt{3}}{6}$ (B) 0 (C) $\frac{\pi+2-3\sqrt{3}}{6}$ (D) π

Q41. JEE Main 2023 (Online) 12th April Morning Shift

If the local maximum value of the function $f(x) = \left(\frac{\sqrt{3e}}{2\sin x}\right)^{\sin^2 x}, x \in \left(0, \frac{\pi}{2}\right)$, is $\frac{k}{e}$, then $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ is equal to

(A) $e^3 + e^6 + e^{10}$ (B) $e^3 + e^5 + e^{11}$ (C) $e^3 + e^6 + e^{11}$ (D) $e^5 + e^6 + e^{11}$

Q42. JEE Main 2023 (Online) 10th April Evening Shift

Let $g(x) = f(x) + f(1 - x)$ and $f''(x) > 0, x \in (0, 1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{-1}(2\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right) + \tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to :

(A) $\frac{3\pi}{4}$ (B) π (C) $\frac{5\pi}{4}$ (D) $\frac{3\pi}{2}$

Q43. JEE Main 2023 (Online) 10th April Morning Shift

A square piece of tin of side 30 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. If the volume of the box is maximum, then its surface area (in cm^2) is equal to :

(A) 1025 (B) 900 (C) 800 (D) 675

Q44. JEE Main 2023 (Online) 1st February Evening Shift

The sum of the absolute maximum and minimum values of the function $f(x) = |x^2 - 5x + 6| - 3x + 2$ in the interval $[-1, 3]$ is equal to :

(A) 13 (B) 24 (C) 10 (D) 12

Q45. JEE Main 2023 (Online) 31st January Morning Shift

A wire of length $20\ \mathrm{m}$ is to be cut into two pieces. A piece of length l_1 is bent to make a square of area A_1 and the other piece of length l_2 is made into a circle of area A_2. If $2A_1 + 3A_2$ is minimum then $(\pi l_1) : l_2$ is equal to :

(A) 6 : 1 (B) 1 : 6 (C) 4 : 1 (D) 3 : 1

Q46. JEE Main 2023 (Online) 30th January Evening Shift

If the functions $f(x) = \frac{x^3}{3} + 2bx + \frac{ax^2}{2}$

and $g(x) = \frac{x^3}{3} + ax + bx^2, a \neq 2b$

have a common extreme point, then $a + 2b + 7$ is equal to :

(A) 6 (B) $\frac{3}{2}$ (C) 3 (D) 4

Q47. JEE Main 2023 (Online) 25th January Evening Shift

Let the function $f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$ have a maxima for some value of $x < 0$ and a minima for some value of $x > 0$. Then, the set of all values of p is

(A) $\left(-\frac{9}{2}, \frac{9}{2}\right)$ (B) $\left(\frac{9}{2}, \infty\right)$

(C) $\left(0, \frac{9}{2}\right)$ (D) $\left(-\infty, \frac{9}{2}\right)$

Q48. JEE Main 2023 (Online) 25th January Morning Shift

Let $x = 2$ be a local minima of the function $f(x) = 2x^4 - 18x^2 + 8x + 12, x \in (-4, 4)$. If M is local maximum value of the function f in $(-4, 4)$, then M =

(A) $18\sqrt{6} - \frac{33}{2}$ (B) $18\sqrt{6} - \frac{31}{2}$ (C) $12\sqrt{6} - \frac{33}{2}$ (D) $12\sqrt{6} - \frac{31}{2}$

Q49. JEE Main 2023 (Online) 25th January Morning Shift

Let $f:(0,1) \to \mathbb{R}$ be a function defined $f(x)=\frac{1}{1-e^{-x}}$, and $g(x)=(f(-x)-f(x))$. Consider two statements

(I) g is an increasing function in (0, 1)

(II) g is one-one in (0, 1)

Then,

(A) Both (I) and (II) are true

(B) Neither (I) nor (II) is true

(C) Only (II) is true

(D) Only (I) is true

Q50. JEE Main 2022 (Online) 29th July Morning Shift

Let $f(x)=3^{\left(x^2-2\right)^3+4}, x \in \mathrm{R}$. Then which of the following statements are true?

$\mathrm{P}: x=0$ is a point of local minima of f

$\mathrm{Q}: x=\sqrt{2}$ is a point of inflection of f

$R: f'$ is increasing for $x>\sqrt{2}$

(A) Only P and Q

(B) Only P and R

(C) Only Q and R

(D) All P, Q and R

Q51. JEE Main 2022 (Online) 28th July Evening Shift

The function $f(x)=x e^{x(1-x)}, x \in \mathbb{R}$, is :

(A) increasing in $\left(-\frac{1}{2}, 1\right)$

(B) decreasing in $\left(\frac{1}{2}, 2\right)$

(C) increasing in $\left(-1,-\frac{1}{2}\right)$

(D) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Q52. JEE Main 2022 (Online) 28th July Morning Shift

If the minimum value of $f(x)=\frac{5x^2}{2}+\frac{\alpha}{x^5}, x>0$, is 14 , then the value of α is equal to :

(A) 32

(B) 64

(C) 128

(D) 256

Q53. JEE Main 2022 (Online) 26th July Evening Shift

If the maximum value of a, for which the function $f_a(x)=\tan^{-1} 2x-3ax+7$ is non-decreasing in $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$, is $\bar{a}$, then $f_{\bar{a}}\left(\frac{\pi}{8}\right)$ is equal to :

(A) $8-\frac{9\pi}{4(9+\pi^2)}$

(B) $8-\frac{4\pi}{9(4+\pi^2)}$

(C) $8\left(\frac{1+\pi^2}{9+\pi^2}\right)$

(D) $8-\frac{\pi}{4}$

Q54. JEE Main 2022 (Online) 26th July Morning Shift

Let $f(x)=\begin{cases}x^3-x^2+10x-7, & x\le 1\\ -2x+\log_2(b^2-4), & x>1\end{cases}$.

Then the set of all values of b, for which f(x) has maximum value at x = 1, is :

(A) $(-6,-2)$ (B) $(2,6)$

(C) $[-6,-2)\cup(2,6]$ (D) $\left[-\sqrt{6},-2\right)\cup\left(2,\sqrt{6}\right]$

Q55. JEE Main 2022 (Online) 25th July Morning Shift

If the absolute maximum value of the function $f(x)=\left(x^2-2x+7\right)e^{\left(4x^3-12x^2-180x+31\right)}$ in the interval $[-3,0]$ is $f(\alpha)$, then :

(A) $\alpha=0$ (B) $\alpha=-3$ (C) $\alpha\in(-1,0)$ (D) $\alpha\in(-3,-1]$

Q56. JEE Main 2022 (Online) 25th July Morning Shift

The curve $y(x)=ax^3+bx^2+cx+5$ touches the x-axis at the point $P(-2,0)$ and cuts the y-axis at the point Q, where y' is equal to 3 . Then the local maximum value of $y(x)$ is:

(A) $\frac{27}{4}$ (B) $\frac{29}{4}$ (C) $\frac{37}{4}$ (D) $\frac{9}{2}$

Q57. JEE Main 2022 (Online) 30th June Morning Shift

If xy^4 attains maximum value at the point (x, y) on the line passing through the points (50 + α, 0) and (0, 50 + α), α > 0, then (x, y) also lies on the line :

(A) y = 4x (B) x = 4y (C) y = 4x + α (D) x = 4y $-\alpha$

Q58. JEE Main 2022 (Online) 30th June Morning Shift

Let $f(x)=4x^3-11x^2+8x-5,\ x\in R$. Then f :

(A) has a local minima at $x=\frac{1}{2}$ (B) has a local minima at $x=\frac{3}{4}$

(C) is increasing in $\left(\frac{1}{2},\frac{3}{4}\right)$ (D) is decreasing in $\left(\frac{1}{2},\frac{4}{3}\right)$

Q59. JEE Main 2022 (Online) 29th June Morning Shift

A wire of length 22 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into an equilateral triangle. Then, the length of the side of the equilateral triangle, so that the combined area of the square and the equilateral triangle is minimum, is :

(A) $\frac{22}{9+4\sqrt{3}}$ (B) $\frac{66}{9+4\sqrt{3}}$ (C) $\frac{22}{4+9\sqrt{3}}$ (D) $\frac{66}{4+9\sqrt{3}}$

Q60. JEE Main 2022 (Online) 28th June Morning Shift

The number of real solutions of

$x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____.

(A) 0 (B) 1 (C) 3 (D) 5

Q61. JEE Main 2022 (Online) 26th June Evening Shift

Consider a cuboid of sides 2x, 4x and 5x and a closed hemisphere of radius r. If the sum of their surface areas is a constant k, then the ratio x : r, for which the sum of their volumes is maximum, is :

(A) 2 : 5 (B) 19 : 45 (C) 3 : 8 (D) 19 : 15

Q62. JEE Main 2022 (Online) 26th June Morning Shift

The sum of the absolute minimum and the absolute maximum values of the

function f(x) = $|3x - x^2 + 2| - x$ in the interval $[-1, 2]$ is :

(A) $\frac{\sqrt{17}+3}{2}$ (B) $\frac{\sqrt{17}+5}{2}$ (C) 5 (D) $\frac{9-\sqrt{17}}{2}$

Q63. JEE Main 2022 (Online) 26th June Morning Shift

Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$. If [a, b] is the range of the function f, then 4a $-$ b is equal to :

(A) 11 (B) $11 - \pi$ (C) $11 + \pi$ (D) $15 - \pi$

Q64. JEE Main 2022 (Online) 25th June Evening Shift

Water is being filled at the rate of 1 cm^3 / sec in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is

(A) 5 (B) $\frac{\sqrt{21}}{5}$ (C) $\frac{\sqrt{26}}{5}$ (D) $\frac{\sqrt{26}}{10}$

Q65. JEE Main 2022 (Online) 24th June Evening Shift

Let λ^* be the largest value of λ for which the function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is increasing for all x $\in$ R. Then $f_{\lambda^*}(1) + f_{\lambda^*}(-1)$ is equal to :

(A) 36 (B) 48 (C) 64 (D) 72

Q66. JEE Main 2022 (Online) 24th June Morning Shift

The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is :

(A) 9 (B) 10 (C) 11 (D) 12

Q67. JEE Main 2022 (Online) 24th June Morning Shift

For the function

$f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5,\ x > 1$, which one of the following is NOT correct?

(A) f is increasing in (1, 2) and decreasing in (2, ∞)

(B) f(x) = -1 has exactly two solutions

(C) $f'(e) - f''(2) < 0$

(D) f(x) = 0 has a root in the interval (e, e + 1)

Q68. JEE Main 2022 (Online) 24th June Morning Shift

The sum of absolute maximum and absolute minimum values of the function $f(x) = |2x^2 + 3x - 2| + \sin x \cos x$ in the interval [0, 1] is :

(A) $3 + \frac{\sin(1)\cos^2(\frac{1}{2})}{2}$

(B) $3 + \frac{1}{2}(1 + 2\cos(1))\sin(1)$

(C) $5 + \frac{1}{2}(\sin(1) + \sin(2))$

(D) $2 + \sin\left(\frac{1}{2}\right)\cos\left(\frac{1}{2}\right)$

Q69. JEE Main 2021 (Online) 1st September Evening Shift

The function $f(x) = x^3 - 6x^2 + ax + b$ is such that $f(2) = f(4) = 0$. Consider two statements :

Statement 1 : there exists $x_1, x_2 \in (2, 4)$, $x_1 < x_2$, such that $f'(x_1) = -1$ and $f'(x_2) = 0$.

Statement 2 : there exists $x_3, x_4 \in (2, 4)$, $x_3 < x_4$, such that f is decreasing in $(2, x_4)$, increasing in $(x_4, 4)$ and $2f'(x_3) = \sqrt{3}f(x_4)$.

Then

(A) both Statement 1 and Statement 2 are true

(B) Statement 1 is false and Statement 2 is true

(C) both Statement 1 and Statement 2 are false

(D) Statement 1 is true and Statement 2 is false

Q70. JEE Main 2021 (Online) 31st August Morning Shift

The number of real roots of the equation

$e^{4x} + 2e^{3x} - e^x - 6 = 0$ is :

(A) 2 (B) 4 (C) 1 (D) 0

Q71. JEE Main 2021 (Online) 27th August Evening Shift

A box open from top is made from a rectangular sheet of dimension a $\times$ b by cutting squares each of side x from each of the four corners and folding up the flaps. If the volume of the box is maximum, then x is equal to :

(A) $\frac{a+b-\sqrt{a^2+b^2-ab}}{12}$ (B) $\frac{a+b-\sqrt{a^2+b^2+ab}}{6}$

(C) $\frac{a+b-\sqrt{a^2+b^2-ab}}{6}$ (D) $\frac{a+b+\sqrt{a^2+b^2+ab}}{6}$

Q72. JEE Main 2021 (Online) 27th August Morning Shift

A wire of length 20 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a regular hexagon. Then the length of the side (in meters) of the hexagon, so that the combined area of the square and the hexagon is minimum, is :

(A) $\frac{5}{2+\sqrt{3}}$ (B) $\frac{10}{2+3\sqrt{3}}$ (C) $\frac{5}{3+\sqrt{3}}$ (D) $\frac{10}{3+2\sqrt{3}}$

Q73. JEE Main 2021 (Online) 26th August Evening Shift

The local maximum value of the function $f(x)=\left(\frac{2}{x}\right)^{x^2}$, x > 0, is

(A) $\left(2\sqrt{e}\right)^{\frac{1}{e}}$ (B) $\left(\frac{4}{\sqrt{e}}\right)^{\frac{e}{4}}$

(C) $(e)^{\frac{2}{e}}$ (D) 1

Q74. JEE Main 2021 (Online) 25th July Morning Shift

Let $f(x)=3\sin^4 x+10\sin^3 x+6\sin^2 x-3$, $x\in\left[-\frac{\pi}{6},\frac{\pi}{2}\right]$. Then, f is :

(A) increasing in $\left(-\frac{\pi}{6},\frac{\pi}{2}\right)$ (B) decreasing in $\left(0,\frac{\pi}{2}\right)$

(C) increasing in $\left(-\frac{\pi}{6},0\right)$ (D) decreasing in $\left(-\frac{\pi}{6},0\right)$

Q75. JEE Main 2021 (Online) 22th July Evening Shift

Let f : R $\rightarrow$ R be defined as

$f(x)=\begin{cases}-\frac{4}{3}x^3+2x^2+3x, & x>0\\ 3xe^x, & x\le 0\end{cases}$. Then f is increasing function in the interval

(A) $\left(-\frac{1}{2},2\right)$ (B) (0,2)

(C) $\left(-1,\frac{3}{2}\right)$ (D) $(-3,-1)$

Q76. JEE Main 2021 (Online) 20th July Evening Shift

The sum of all the local minimum values of the twice differentiable function f : R $\rightarrow$ R defined by $f(x)=x^3-3x^2-\frac{3f''(2)}{2}x+f''(1)$ is :

(A) -22 (B) 5 (C) -27 (D) 0

Q77. JEE Main 2021 (Online) 20th July Morning Shift

Let $A = [a_{ij}]$ be a 3 × 3 matrix, where $a_{ij} = \begin{cases} 1 & , \quad if\, i = j \\ -x & , \quad if\ |i-j| = 1 \\ 2x+1 & , \quad otherwise. \end{cases}$

Let a function f : R → R be defined as f(x) = det(A). Then the sum of maximum and minimum values of f on R is equal to:

(A) $-\frac{20}{27}$ (B) $\frac{88}{27}$ (C) $\frac{20}{27}$ (D) $-\frac{88}{27}$

Q78. JEE Main 2021 (Online) 20th July Morning Shift

Let 'a' be a real number such that the function f(x) = ax^2 + 6x − 15, x ∈ R is increasing in $(-\infty, \frac{3}{4})$ and decreasing in $(\frac{3}{4}, \infty)$. Then the function g(x) = ax^2 − 6x + 15, x∈R has a :

(A) local maximum at x = $-\frac{3}{4}$

(B) local minimum at x = $-\frac{3}{4}$

(C) local maximum at x = $\frac{3}{4}$

(D) local minimum at x = $\frac{3}{4}$

Q79. JEE Main 2021 (Online) 17th March Evening Shift

Consider the function f : R → R defined by

$f(x) = \begin{cases} \left(2 - \sin\left(\frac{1}{x}\right)\right)|x|, x \neq 0 \\ 0, \ x = 0 \end{cases}$. Then f is :

(A) not monotonic on $(-\infty, 0)$ and $(0, \infty)$

(B) monotonic on $(0, \infty)$ only

(C) monotonic on $(-\infty, 0)$ only

(D) monotonic on $(-\infty, 0) \cup (0, \infty)$

Q80. JEE Main 2021 (Online) 16th March Evening Shift

Let f be a real valued function, defined on R − {−1, 1} and given by

f(x) = 3 $\log_e \left|\frac{x-1}{x+1}\right| - \frac{2}{x-1}$.

Then in which of the following intervals, function f(x) is increasing?

(A) $(-\infty, -1) \cup \left([\frac{1}{2}, \infty) - \{1\}\right)$

(B) $(-\infty, \infty) - \{-1, 1)$

(C) $(-\infty, \frac{1}{2}] - \{-1\}$

(D) $(-1, \frac{1}{2}]$

Q81. JEE Main 2021 (Online) 26th February Morning Shift

The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point :

(A) $\left(3, \frac{21}{2}\right)$ (B) (0, 0)

(C) (2, 9) (D) (2, 2)

Q82. JEE Main 2021 (Online) 24th February Evening Shift

Let $f : R \to R$ be defined as

$$f(x) = \begin{cases} -55x, & if\, x < -5 \\ 2x^3 - 3x^2 - 120x, & if\ -5 \le x \le 4 \\ 2x^3 - 3x^2 - 36x - 336, & if\, x > 4, \end{cases}$$

Let A = {x ∈ R : f is increasing}. Then A is equal to :

(A) $(-5, \infty)$ (B) $(-\infty, -5) \cup (4, \infty)$

(C) $(-5, -4) \cup (4, \infty)$ (D) $(-\infty, -5) \cup (-4, \infty)$

Q83. JEE Main 2021 (Online) 24th February Morning Shift

The function
f(x) = $\frac{4x^3 - 3x^2}{6} - 2\sin x + (2x - 1)\cos x$:

(A) increases in $\left(-\infty, \frac{1}{2}\right]$ (B) decreases in $\left(-\infty, \frac{1}{2}\right]$

(C) increases in $\left[\frac{1}{2}, \infty\right)$ (D) decreases in $\left[\frac{1}{2}, \infty\right)$

Q84. JEE Main 2020 (Online) 6th September Evening Slot

The set of all real values of λ for which the function

$$f(x) = \left(1 - \cos^2 x\right)(\lambda + \sin x), x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

has exactly one maxima and exactly one minima, is :

(A) $\left(-\frac{3}{2}, \frac{3}{2}\right) - \{0\}$ (B) $\left(-\frac{3}{2}, \frac{3}{2}\right)$

(C) $\left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$ (D) $\left(-\frac{1}{2}, \frac{1}{2}\right)$

Q85. JEE Main 2020 (Online) 5th September Evening Slot

If x = 1 is a critical point of the function
f(x) = $(3x^2 + ax - 2 - a)e^x$, then :

(A) x = 1 is a local maxima and x = $-\frac{2}{3}$ is a local minima of f.

(B) x = 1 and x = $-\frac{2}{3}$ are local maxima of f.

(C) x = 1 and x = $-\frac{2}{3}$ are local minima of f.

(D) x = 1 is a local minima and x = $-\frac{2}{3}$ is a local maxima of f.

Q86. JEE Main 2020 (Online) 5th September Morning Slot

If the point P on the curve, $4x^2 + 5y^2 = 20$ is
farthest from the point Q(0, -4), then PQ^2 is equal to:

(A) 36 (B) 48 (C) 21 (D) 29

Q87. JEE Main 2020 (Online) 4th September Evening Slot

The area (in sq. units) of the largest rectangle ABCD whose vertices A and B lie on the x-axis and vertices C and D lie on the parabola, $y = x^2 - 1$ below the x-axis, is :

(A) $\frac{1}{3\sqrt{3}}$ (B) $\frac{2}{3\sqrt{3}}$ (C) $\frac{4}{3\sqrt{3}}$ (D) $\frac{4}{3}$

Q88. JEE Main 2020 (Online) 4th September Morning Slot

Let f be a twice differentiable function on (1, 6). If f(2) = 8, f'(2) = 5, $f'(x) \geq 1$ and $f''(x) \geq 4$, for all $x \in (1, 6)$, then :

(A) $f(5) \leq 10$ (B) $f(5) + f'(5) \geq 28$
(C) $f(5) + f'(5) \leq 26$ (D) $f'(5) + f''(5) \leq 20$

Q89. JEE Main 2020 (Online) 3rd September Evening Slot

Suppose f(x) is a polynomial of degree four, having critical points at −1, 0, 1. If
$T = \{x \in R \mid f(x) = f(0)\}$, then the sum of squares of all the elements of T is :

(A) 6 (B) 2 (C) 8 (D) 4

Q90. JEE Main 2020 (Online) 3rd September Evening Slot

If the surface area of a cube is increasing at a rate of 3.6 cm^2/sec, retaining its shape; then the rate of change of its volume (in cm^3/sec), when the length of a side of the cube is 10 cm, is :

(A) 9 (B) 10 (C) 18 (D) 20

Q91. JEE Main 2020 (Online) 3rd September Morning Slot

The function, $f(x) = (3x - 7)x^{2/3}$, $x \in R$, is increasing for all x lying in :

(A) $(-\infty, 0) \cup \left(\frac{3}{7}, \infty\right)$ (B) $(-\infty, 0) \cup \left(\frac{14}{15}, \infty\right)$
(C) $\left(-\infty, \frac{14}{15}\right)$ (D) $\left(-\infty, \frac{14}{15}\right) \cup (0, \infty)$

Q92. JEE Main 2020 (Online) 2nd September Evening Slot

Let f : $(-1, \infty) \to R$ be defined by f(0) = 1 and
f(x) = $\frac{1}{x}\log_e(1+x)$, $x \neq 0$. Then the function f :

(A) decreases in $(-1, \infty)$

(B) decreases in $(-1, 0)$ and increases in $(0, \infty)$

(C) increases in $(-1, \infty)$

(D) increases in $(-1, 0)$ and decreases in $(0, \infty)$

Q93. JEE Main 2020 (Online) 2nd September Morning Slot

If p(x) be a polynomial of degree three that has a local maximum value 8 at x = 1 and a local minimum value 4 at x = 2; then p(0) is equal to :

(A) 6 (B) 12 (C) -12 (D) -24

Q94. JEE Main 2020 (Online) 9th January Morning Slot

A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^3/min. When the thickness of ice is 5 cm, then the rate (in cm/min.) at which the thickness of ice decreases, is :

(A) $\frac{1}{18\pi}$ (B) $\frac{1}{36\pi}$ (C) $\frac{1}{54\pi}$ (D) $\frac{5}{6\pi}$

Q95. JEE Main 2020 (Online) 8th January Morning Slot

Let ƒ(x) = $x\cos^{-1}(-\sin|x|)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then which of the following is true?

(A) ƒ' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$

(B) ƒ '(0) = $-\frac{\pi}{2}$

(C) ƒ is not differentiable at x = 0

(D) ƒ' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

Q96. JEE Main 2020 (Online) 7th January Evening Slot

Let ƒ(x) be a polynomial of degree 5 such that x = ±1 are its critical points.

If $\lim_{x\to 0}\left(2 + \frac{f(x)}{x^3}\right) = 4$, then which one of the following is not true?

(A) ƒ(1) - 4ƒ(-1) = 4.

(B) x = 1 is a point of minima and x = -1 is a point of maxima of ƒ.

(C) x = 1 is a point of maxima and x = -1 is a point of minimum of ƒ.

(D) ƒ is an odd function.

Q97. JEE Main 2019 (Online) 12th April Morning Slot

If m is the minimum value of k for which the function f(x) = $x\sqrt{kx - x^2}$ is increasing in the interval [0,3] and M is the maximum value of f in [0, 3] when k = m, then the ordered pair (m, M) is equal to :

(A) $\left(5, 3\sqrt{6}\right)$ (B) $\left(4, 3\sqrt{3}\right)$ (C) $\left(4, 3\sqrt{2}\right)$ (D) $\left(3, 3\sqrt{3}\right)$

Q98. JEE Main 2019 (Online) 12th April Morning Slot

A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec, then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is :

(A) $\frac{25}{3}$ (B) 25 (C) $25\sqrt{3}$ (D) $\frac{25}{\sqrt{3}}$

Q99. JEE Main 2019 (Online) 10th April Evening Slot

A spherical iron ball of radius 10 cm is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^3 /min. When the thickness of the ice is 5 cm, then the rate at which the thickness (in cm/min) of the ice decreases, is :

(A) $\frac{5}{6\pi}$ (B) $\frac{1}{9\pi}$ (C) $\frac{1}{36\pi}$ (D) $\frac{1}{18\pi}$

Q100. JEE Main 2019 (Online) 9th April Evening Slot

A water tank has the shape of an inverted right circular cone, whose semi-vertical angle is $\tan^{-1}\left(\frac{1}{2}\right)$. Water is poured into it at a constant rate of 5 cubic meter per minute. Then the rate (in m/min.), at which the level of water is rising at the instant when the depth of water in the tank is 10m; is :-

(A) $\frac{1}{15\pi}$ (B) $\frac{1}{5\pi}$ (C) $\frac{1}{10\pi}$ (D) $\frac{2}{\pi}$

Q101. JEE Main 2019 (Online) 9th April Morning Slot

If ƒ(x) is a non-zero polynomial of degree four, having local extreme points at x = –1, 0, 1; then the set

S = {x ∈ R : ƒ(x) = ƒ(0)}

Contains exactly :

(A) four rational numbers. (B) four irrational numbers.

(C) two irrational and one rational number. (D) two irrational and two rational numbes.

Q102. JEE Main 2019 (Online) 8th April Evening Slot

The height of a right circular cylinder of maximum volume inscribed in a sphere of radius 3 is

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\sqrt{6}$ (D) $\frac{2}{3}\sqrt{3}$

Q103. JEE Main 2019 (Online) 8th April Morning Slot

If S_1 and S_2 are respectively the sets of local minimum and local maximum points of the function,

$f(x) = 9x^4 + 12x^3 - 36x^2 + 25, x \in R$, then :

(A) $S_1 = \{-1\}; S_2 = \{0, 2\}$

(B) $S_1 = \{-2\}; S_2 = \{0, 1\}$

(C) $S_1 = \{-2, 0\}; S_2 = \{1\}$

(D) $S_1 = \{-2, 1\}; S_2 = \{0\}$

Q104. JEE Main 2019 (Online) 8th April Morning Slot

Let $f : [0, 2] \to R$ be a twice differentiable function such that $f''(x) > 0$, for all $x \in (0, 2)$. If $\phi(x) = f(x) + f(2 - x)$, then ϕ is :

(A) decreasing on (0, 2)

(B) decreasing on (0, 1) and increasing on (1, 2)

(C) increasing on (0, 2)

(D) increasing on (0, 1) and decreasing on (1, 2)

Q105. JEE Main 2019 (Online) 12th January Evening Slot

If the function f given by $f(x) = x^3 - 3(a - 2)x^2 + 3ax + 7$, for some $a \in R$ is increasing in (0, 1] and decreasing in [1, 5), then a root of the equation, $\frac{f(x)-14}{(x-1)^2} = 0\,(x \neq 1)$ is :

(A) -7 (B) 5 (C) 7 (D) 6

Q106. JEE Main 2019 (Online) 11th January Evening Slot

Let $f(x) = \frac{x}{\sqrt{a^2+x^2}} - \frac{d-x}{\sqrt{b^2+(d-x)^2}}$, $x \in R$, where a, b and d are non-zero real constants. Then :

(A) f is an increasing function of x

(B) f is neither increasing nor decreasing function of x

(C) f ' is not a continuous function of x

(D) f is a decreasing function of x

Q107. JEE Main 2019 (Online) 11th January Morning Slot

The maximum value of the function $f(x) = 3x^3 - 18x^2 + 27x - 40$ on the set $S = \{x \in R : x^2 + 30 \leq 11x\}$ is :

(A) -222 (B) -122 (C) 122 (D) 222

Q108. JEE Main 2019 (Online) 10th January Evening Slot

A helicopter is flying along the curve given by $y - x^{3/2} = 7$, $(x \geq 0)$. A soldier positioned at the point $\left(\frac{1}{2}, 7\right)$ wants to shoot down the helicopter when it is nearest to him. Then this nearest distance is -

(A) $\frac{1}{6}\sqrt{\frac{7}{3}}$ (B) $\frac{\sqrt{5}}{6}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}\sqrt{\frac{7}{3}}$

Q109. JEE Main 2019 (Online) 10th January Morning Slot

The shortest distance between the point $\left(\frac{3}{2}, 0\right)$ and the curve $y = \sqrt{x}$, $(x > 0)$, is -

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{5}{4}$ (C) $\frac{3}{2}$ (D) $\frac{\sqrt{5}}{2}$

Q110. JEE Main 2019 (Online) 9th January Morning Slot

The maximum volume (in cu.m) of the right circular cone having slant height 3 m is :

(A) $2\sqrt{3}\pi$ (B) $3\sqrt{3}\pi$ (C) 6π (D) $\frac{4}{3}\pi$

Q111. JEE Main 2018 (Online) 16th April Morning Slot

Let M and m be respectively the absolute maximum and the absolute minimum values of the function, $f(x) = 2x^3 - 9x^2 + 12x + 5$ in the interval [0, 3]. Then $M - m$ is equal to :

(A) 5 (B) 9 (C) 4 (D) 1

Q112. JEE Main 2018 (Offline)

Let $f(x) = x^2 + \frac{1}{x^2}$ and $g(x) = x - \frac{1}{x}$,
$x \in R - \{-1, 0, 1\}$.
If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of h(x) is

(A) $2\sqrt{2}$ (B) 3 (C) -3 (D) $-2\sqrt{2}$

Q113. JEE Main 2018 (Online) 15th April Morning Slot

If a right circular cone, having maximum volume, is inscribed in a sphere of radius 3 cm, then the curved surface area (in cm^2) of this cone is :

(A) $6\sqrt{2}\pi$ (B) $6\sqrt{3}\pi$ (C) $8\sqrt{2}\pi$ (D) $8\sqrt{3}\pi$

Q114. JEE Main 2017 (Online) 9th April Morning Slot

The function f defined by

$f(x) = x^3 - 3x^2 + 5x + 7$, is :

(A) increasing in **R**.

(B) decreasing in **R**.

(C) decreasing in $(0, \infty)$ and increasing in $(-\infty, 0)$

(D) increasing in $(0, \infty)$ and decreasing in $(-\infty, 0)$

Q115. JEE Main 2017 (Offline)

Twenty meters of wire is available for fencing off a flower-bed in the form of a circular sector. Then the maximum area (in sq. m) of the flower-bed, is :

(A) 10 (B) 25 (C) 30 (D) 12.5

Q116. JEE Main 2016 (Online) 10th April Morning Slot

Let $f(x) = \sin^4 x + \cos^4 x$. Then f is an increasing function in the interval :

(A) $]0, \frac{\pi}{4}[$ (B) $]\frac{\pi}{4}, \frac{\pi}{2}[$

(C) $]\frac{\pi}{2}, \frac{5\pi}{8}[$ (D) $]\frac{5\pi}{8}, \frac{3\pi}{4}[$

Q117. JEE Main 2016 (Online) 9th April Morning Slot

The minimum distance of a point on the curve $y = x^2 - 4$ from the origin is :

(A) $\frac{\sqrt{19}}{2}$ (B) $\sqrt{\frac{15}{2}}$ (C) $\frac{\sqrt{15}}{2}$ (D) $\sqrt{\frac{19}{2}}$

Q118. JEE Main 2016 (Offline)

A wire of length 2 units is cut into two parts which are bent respectively to form a square of side $= x$ units and a circle of radius $= r$ units. If the sum of the areas of the square and the circle so formed is minimum, then:

(A) $x = 2r$ (B) $2x = r$

(C) $2x = (\pi + 4)r$ (D) $(4 - \pi)x = \pi\, r$

Q119. JEE Main 2015 (Offline)

Let $f(x)$ be a polynomial of degree four having extreme values at $x = 1$ and $x = 2$. If $\lim\limits_{x \to 0} \left[1 + \frac{f(x)}{x^2}\right] = 3$, then $f(2)$ is equal to :

(A) 0 (B) 4 (C) -8 (D) -4

Q120. JEE Main 2014 (Offline)

If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$ then

(A) $\alpha = 2, \beta = -\frac{1}{2}$

(B) $\alpha = 2, \beta = \frac{1}{2}$

(C) $\alpha = -6, \beta = \frac{1}{2}$

(D) $\alpha = -6, \beta = -\frac{1}{2}$

Q121. JEE Main 2013 (Offline)

The real number k for which the equation, $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$

(A) lies between 1 and 2

(B) lies between 2 and 3

(C) lies between -1 and 0

(D) does not exist.

Q122. AIEEE 2012

A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is :

(A) $\frac{9}{7}$ (B) $\frac{7}{9}$ (C) $\frac{2}{9}$ (D) $\frac{9}{2}$

Q123. AIEEE 2012

Let $a, b \in R$ be such that the function f given by $f(x) = In|x| + bx^2 + ax,\ x \neq 0$ has extreme values at $x = -1$ and $x = 2$

Statement-1 : f has local maximum at $x = -1$ and at $x = 2$.

Statement-2 : $a = \frac{1}{2}$ and $b = \frac{-1}{4}$

(A) Statement - 1 is false, Statement - 2 is true.

(B) Statement - 1 is true , Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1.

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** a correct explanation for Statement - 1.

(D) Statement - 1 is true, Statement - 2 is false.

Q124. AIEEE 2012

A line is drawn through the point $(1, 2)$ to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. If the area of the triangle OPQ is least, then the slope of the line PQ is :

(A) $-\frac{1}{4}$ (B) -4 (C) -2 (D) $-\frac{1}{2}$

Q125. AIEEE 2011

For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t} \sin t \, dt$. Then f has

(A) local minimum at π and 2π

(B) local minimum at π and local maximum at 2π

(C) local maximum at π and local minimum at 2π

(D) local maximum at π and 2π

Q126. AIEEE 2010

Let $f : R \to R$ be defined by

$$f(x) = \begin{cases} k - 2x, & if \quad x \leq -1 \\ 2x + 3, & if \quad x > -1 \end{cases}$$

If $f$has a local minimum at $x = -1$, then a possible value of k is

(A) 0 (B) $-\frac{1}{2}$ (C) -1 (D) 1

Q127. AIEEE 2010

Let $f : R \to R$ be a continuous function defined by

$$f(x) = \frac{1}{e^x + 2e^{-x}}$$

Statement - 1 : $f(c) = \frac{1}{3}$, for some $c \in R$.

Statement - 2 : $0 < f(x) \leq \frac{1}{2\sqrt{2}}$, for all $x \in R$

(A) Statement - 1 is true, Statement -2 is true; Statement - 2 is **not** a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement - 2 is false.

(C) Statement - 1 is false, Statement - 2 is true.

(D) Statement - 1 is true, Statement -2 is true; Statement -2 is a correct explanation for Statement - 1.

Q128. AIEEE 2009

Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$:

(A) $P(-1)$ is not minimum but $P(1)$ is the maximum of P

(B) $P(-1)$ is the minimum but $P(1)$ is not the maximum of P

(C) Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P

(D) $P(-1)$ is the minimum and $P(1)$ is the maximum of P

Q129. AIEEE 2008

Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then which one of the following holds?

(A) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$

(B) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$

(C) The cubic has minima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(D) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

Q130. AIEEE 2008

How many real solutions does the equation $x^7 + 14x^5 + 16x^3 + 30x - 560 = 0$ have?

(A) 7 (B) 1 (C) 3 (D) 5

Q131. AIEEE 2007

The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(C) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (D) $\left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

Q132. AIEEE 2007

If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{2}$ (D) 2

Q133. AIEEE 2006

The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minimum at

(A) $x = 2$ (B) $x = -2$ (C) $x = 0$ (D) $x = 1$

Q134. AIEEE 2006

A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x. The maximum area enclosed by the park is

(A) $\frac{3}{2}x^2$ (B) $\sqrt{\frac{x^3}{8}}$ (C) $\frac{1}{2}x^2$ (D) πx^2

Q135. AIEEE 2005

Area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(A) $2ab$ (B) ab (C) $\sqrt{ab}$ (D) $\frac{a}{b}$

Q136. AIEEE 2005

A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^3 /min. When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases is

(A) $\frac{1}{36\pi}$ cm/min (B) $\frac{1}{18\pi}$ cm/min (C) $\frac{1}{54\pi}$ cm/min (D) $\frac{5}{6\pi}$ cm/min

Q137. AIEEE 2005

A function is matched below against an interval where it is supposed to be increasing. Which of the following pairs is incorrectly matched?

(A)

Interval	Function
$(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$

(B)

Interval	Function
$[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$

(C)

Interval	Function
$(-\infty, \frac{1}{3}]$	$3x^2 - 2x + 1$

(D)

Interval	Function
$(-\infty, -4)$	$x^3 + 6x^2 + 6$

Q138. AIEEE 2005

A lizard, at an initial distance of 21 cm behind an insect moves from rest with an acceleration of $2\ \text{cm/s}^2$ and pursues the insect which is crawling uniformly along a straight line at a speed of $20\ \text{cm/s}$. Then the lizard will catch the insect after :

(A) 20 s (B) 1 s (C) 21 s (D) 24 s

Q139. AIEEE 2004

A point on the parabola $y^2 = 18x$ at which the ordinate increases at twice the rate of the abscissa is

(A) $(\frac{9}{8}, \frac{9}{2})$ (B) $(2, -4)$
(C) $(\frac{-9}{8}, \frac{9}{2})$ (D) $(2, 4)$

Q140. AIEEE 2003

The real number x when added to its inverse gives the minimum sum at x equal :

(A) -2 (B) 2 (C) 1 (D) -1

Q141. AIEEE 2003

If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals

(A) $\frac{1}{2}$ (B) 3 (C) 1 (D) 2

Q142. AIEEE 2002

The maximum distance from origin of a point on the curve
$x = a\sin t - b\sin\left(\frac{at}{b}\right)$
$y = a\cos t - b\cos\left(\frac{at}{b}\right)$, both $a, b > 0$ is

(A) $a - b$ (B) $a + b$ (C) $\sqrt{a^2 + b^2}$ (D) $\sqrt{a^2 - b^2}$

Numerical 27 questions

Q143. JEE Main 2026 (Online) 24th January Morning Shift

Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t) = \frac{|t+1|}{t^2}, t < 0$, is strictly decreasing. Then the local maximum value of the function $g(x) = 2\log_e(x-2) + \alpha x^2 + 4x - \alpha, x > 2$, is ____

Q144. JEE Main 2026 (Online) 21st January Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that the quadratic equation $f(x)\mathrm{m}^2 - 2f'(x)\mathrm{m} + f''(x) = 0$ in m, has two equal roots for every $x \in \mathbf{R}$. If $f(0) = 1, f'(0) = 2$, and (α, β) is the largest interval in which the function $f(\log_e x - x)$ is increasing, then $\alpha + \beta$ is equal to ____.

Q145. JEE Main 2025 (Online) 2nd April Evening Shift

Let $\mathrm{A}(4, -2), \mathrm{B}(1, 1)$ and $\mathrm{C}(9, -3)$ be the vertices of a triangle ABC. Then the maximum area of the parallelogram AFDE, formed with vertices D, E and F on the sides BC, CA and AB of the triangle ABC respectively, is_____

Q146. JEE Main 2025 (Online) 23rd January Morning Shift

If the set of all values of a, for which the equation $5x^3 - 15x - a = 0$ has three distinct real roots, is the interval (α, β), then $\beta - 2\alpha$ is equal to _____.

Q147. JEE Main 2024 (Online) 9th April Evening Shift

Let the set of all values of p, for which $f(x) = \left(p^2 - 6p + 8\right)\left(\sin^2 2x - \cos^2 2x\right) + 2(2-p)x + 7$ does not have any critical point, be the interval (a, b). Then $16ab$ is equal to _____.

Q148. JEE Main 2024 (Online) 9th April Morning Shift

Let the set of all positive values of λ, for which the point of local minimum of the function $(1+x(\lambda^2-x^2))$ satisfies $\frac{x^2+x+2}{x^2+5x+6}<0$, be (α,β). Then $\alpha^2+\beta^2$ is equal to _____.

Q149. JEE Main 2024 (Online) 8th April Evening Shift

Let A be the region enclosed by the parabola $y^2=2x$ and the line $x=24$. Then the maximum area of the rectangle inscribed in the region A is _____.

Q150. JEE Main 2024 (Online) 5th April Evening Shift

Let the maximum and minimum values of $\left(\sqrt{8x-x^2-12}-4\right)^2+(x-7)^2, x\in\mathbf{R}$ be M and m, respectively. Then $\mathrm{M}^2-\mathrm{m}^2$ is equal to _____.

Q151. JEE Main 2024 (Online) 29th January Morning Shift

Let $f(x)=2^x-x^2, x\in\mathbb{R}$. If m and n are respectively the number of points at which the curves $y=f(x)$ and $y=f'(x)$ intersect the x-axis, then the value of $\mathrm{m}+\mathrm{n}$ is _____.

Q152. JEE Main 2024 (Online) 27th January Morning Shift

Let for a differentiable function $f:(0,\infty)\to\mathbf{R}$, $f(x)-f(y)\geqslant\log_e\left(\frac{x}{y}\right)+x-y, \forall x,y\in(0,\infty)$. Then $\sum\limits_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ is equal to _____.

Q153. JEE Main 2023 (Online) 15th April Morning Shift

Consider the triangles with vertices $A(2,1), B(0,0)$ and $C(t,4), t\in[0,4]$.

If the maximum and the minimum perimeters of such triangles are obtained at

$t=\alpha$ and $t=\beta$ respectively, then $6\alpha+21\beta$ is equal to _____.

Q154. JEE Main 2023 (Online) 8th April Morning Shift

If a_α is the greatest term in the sequence $a_n=\frac{n^3}{n^4+147}, n=1,2,3,\ldots$, then α is equal to _____.

Q155. JEE Main 2023 (Online) 6th April Evening Shift

The number of points, where the curve $y = x^5 - 20x^3 + 50x + 2$ crosses the x-axis, is _____.

Q156. JEE Main 2022 (Online) 27th July Evening Shift

A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____.

Q157. JEE Main 2022 (Online) 26th July Morning Shift

Let the function $f(x) = 2x^2 - \log_e x, x > 0$, be decreasing in $(0, a)$ and increasing in $(a, 4)$. A tangent to the parabola $y^2 = 4ax$ at a point P on it passes through the point $(8a, 8a - 1)$ but does not pass through the point $\left(-\frac{1}{a}, 0\right)$. If the equation of the normal at P is : $\frac{x}{\alpha} + \frac{y}{\beta} = 1$, then $\alpha + \beta$ is equal to _____.

Q158. JEE Main 2022 (Online) 25th July Evening Shift

The sum of the maximum and minimum values of the function $f(x) = |5x - 7| + \left[x^2 + 2x\right]$ in the interval $\left[\frac{5}{4}, 2\right]$, where $[t]$ is the greatest integer $\leq t$, is _____.

Q159. JEE Main 2022 (Online) 29th June Evening Shift

Let f and g be twice differentiable even functions on $(-2, 2)$ such that $f\left(\frac{1}{4}\right) = 0, f\left(\frac{1}{2}\right) = 0, f(1) = 1$ and $g\left(\frac{3}{4}\right) = 0, g(1) = 2$. Then, the minimum number of solutions of $f(x)g''(x) + f'(x)g'(x) = 0$ in $(-2, 2)$ is equal to _____.

Q160. JEE Main 2022 (Online) 25th June Evening Shift

Let $f(x) = |(x - 1)(x^2 - 2x - 3)| + x - 3, x \in R$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval (0, 4), then m + M is equal to _____.

Q161. JEE Main 2021 (Online) 31st August Evening Shift

Let f(x) be a cubic polynomial with f(1) = -10, f(-1) = 6, and has a local minima at x = 1, and f'(x) has a local minima at x = -1. Then f(3) is equal to _____.

Q162. JEE Main 2021 (Online) 31st August Morning Shift

If 'R' is the least value of 'a' such that the function f(x) = x^2 + ax + 1 is increasing on [1, 2] and 'S' is the greatest value of 'a' such that the function f(x) = x^2 + ax + 1 is decreasing on [1, 2], then the value of |R − S| is _____.

Q163. JEE Main 2021 (Online) 27th August Morning Shift

The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is _____.

Q164. JEE Main 2021 (Online) 26th August Morning Shift

A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is k (meter), then $\left(\frac{4}{\pi}+1\right)k$ is equal to _____.

Q165. JEE Main 2021 (Online) 17th March Evening Shift

Let f : $[-1, 1] \to$ R be defined as f(x) = ax^2 + bx + c for all x$\in[-1, 1]$, where a, b, c$\in$R such that f(-1) = 2, f'(-1) = 1 for x$\in(-1, 1)$ the maximum value of f "(x) is $\frac{1}{2}$. If f(x) $\leq \alpha$, x$\in[-1, 1]$, then the least value of α is equal to _____.

Q166. JEE Main 2021 (Online) 26th February Evening Shift

Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval (a, a + 1). Then, |a| is equal to _____.

Q167. JEE Main 2021 (Online) 25th February Morning Shift

Let f(x) be a polynomial of degree 6 in x, in which the coefficient of x^6 is unity and it has extrema at x = -1 and x = 1. If $\lim\limits_{x\to 0}\frac{f(x)}{x^3}=1$, then $5.f(2)$ is equal to _____.

Q168. JEE Main 2021 (Online) 24th February Morning Shift

The minimum value of α for which the equation $\frac{4}{\sin x}+\frac{1}{1-\sin x}=\alpha$ has at least one solution in $\left(0, \frac{\pi}{2}\right)$ is .....

Q169. JEE Main 2020 (Online) 8th January Evening Slot

Let ƒ(x) be a polynomial of degree 3 such that ƒ(−1) = 10, ƒ(1) = −6, ƒ(x) has a critical point at x = −1 and ƒ'(x) has a critical point at x = 1. Then ƒ(x) has a local minima at x = _____.

Out of syllabus 60 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q170. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

Let $f : [2, 4] \to \mathbb{R}$ be a differentiable function such that $(x \log_e x) f'(x) + (\log_e x) f(x) + f(x) \geq 1, x \in [2, 4]$ with $f(2) = \frac{1}{2}$ and $f(4) = \frac{1}{4}$.

Consider the following two statements :

(A) : $f(x) \leq 1$, for all $x \in [2, 4]$

(B) : $f(x) \geq \frac{1}{8}$, for all $x \in [2, 4]$

Then,

(A) Neither statement (A) nor statement (B) is true

(B) Only statement (A) is true

(C) Only statement (B) is true

(D) Both the statements (A) and (B) are true

Q171. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is :

(A) 2 (B) 3 (C) 4 (D) 0

Q172. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let f : R $\to$ R be a function defined by f(x) = $(x - 3)^{n_1} (x - 5)^{n_2}$, $n_1, n_2 \in$ N. Then, which of the following is NOT true?

(A) For n_1 = 3, n_2 = 4, there exists $\alpha \in$ (3, 5) where f attains local maxima.

(B) For n_1 = 4, n_2 = 3, there exists $\alpha \in$ (3, 5) where f attains local minima.

(C) For n_1 = 3, n_2 = 5, there exists $\alpha \in$ (3, 5) where f attains local maxima.

(D) For n_1 = 4, n_2 = 6, there exists $\alpha \in$ (3, 5) where f attains local maxima.

Q173. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let S be the set of all the natural numbers, for which the line $\frac{x}{a} + \frac{y}{b} = 2$ is a tangent to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ at the point (a, b), ab $\neq$ 0. Then :

(A) S = ϕ (B) n(S) = 1

(C) S = {2k : k $\in$ N} (D) S = N

Q174. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

If the angle made by the tangent at the point (x_0, y_0) on the curve $x = 12(t + \sin t \cos t)$, $y = 12(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to:

(A) $6\left(3 + 2\sqrt{2}\right)$ (B) $3\left(7 + 4\sqrt{3}\right)$ (C) 27 (D) 48

Q175. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

If the tangent at the point (x_1, y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve :

(A) $x^2 + \frac{y^2}{81} = 2$ (B) $\frac{y^2}{9} - x^2 = 8$ (C) $y = 4x^2 + 5$ (D) $\frac{x}{3} - y^2 = 2$

Q176. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

Let f be any continuous function on [0, 2] and twice differentiable on (0, 2). If f(0) = 0, f(1) = 1 and f(2) = 2, then

(A) f"(x) = 0 for all $x \in (0, 2)$ (B) f"(x) = 0 for some $x \in (0, 2)$

(C) f'(x) = 0 for some $x \in [0, 2]$ (D) f"(x) > 0 for all $x \in (0, 2)$

Q177. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

Let slope of the tangent line to a curve at any point P(x, y) be given by $\frac{xy^2+y}{x}$. If the curve intersects the line x + 2y = 4 at $x = -2$, then the value of y, for which the point (3, y) lies on the curve, is :

(A) $-\frac{18}{19}$ (B) $-\frac{4}{3}$ (C) $\frac{18}{35}$ (D) $-\frac{18}{11}$

Q178. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

Let f be any function defined on R and let it satisfy the condition :
$|f(x) - f(y)| \leq |(x - y)^2|, \forall (x, y) \in R$

If f(0) = 1, then :

(A) f(x) can take any value in R (B) $f(x) < 0, \forall x \in R$

(C) $f(x) > 0, \forall x \in R$ (D) $f(x) = 0, \forall x \in R$

Q179. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

If the curves, $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{c} + \frac{y^2}{d} = 1$ intersect each other at an angle of 90°, then which of the following relations is TRUE?

(A) a − c = b + d (B) a + b = c + d (C) $ab = \frac{c+d}{a+b}$ (D) a − b = c − d

Q180. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

If Rolle's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$, $x \in [1, 2]$ with $f'\left(\frac{4}{3}\right) = 0$, then ordered pair (a, b) is equal to :

(A) $(-5, -8)$ (B) $(5, -8)$ (C) $(-5, 8)$ (D) $(5, 8)$

Q181. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

For which of the following curves, the line $x + \sqrt{3}y = 2\sqrt{3}$ is the tangent at the point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$?

(A) $2x^2 - 18y^2 = 9$ (B) $y^2 = \frac{1}{6\sqrt{3}}x$ (C) $x^2 + 9y^2 = 9$ (D) $x^2 + y^2 = 7$

Q182. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

If the curve $y = ax^2 + bx + c$, $x \in R$, passes through the point (1, 2) and the tangent line to this curve at origin is y = x, then the possible values of a, b, c are :

(A) a = $-$ 1, b = 1, c = 1 (B) a = 1, b = 1, c = 0

(C) a = $\frac{1}{2}$, b = $\frac{1}{2}$, c = 1 (D) a = 1, b = 0, c = 1

Q183. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

If the tangent to the curve $y = x^3$ at the point $P(t, t^3)$ meets the curve again at Q, then the ordinate of the point which divides PQ internally in the ratio 1 : 2 is :

(A) 0 (B) $2t^3$ (C) $-2t^3$ (D) $-t^3$

Q184. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

If the tangent to the curve, $y = f(x) = x\log_e x$,
$(x > 0)$ at a point (c, f(c)) is parallel to the line-segment
joining the points (1, 0) and (e, e), then c is equal to :

(A) $\frac{e-1}{e}$ (B) $e^{\left(\frac{1}{1-e}\right)}$

(C) $e^{\left(\frac{1}{e-1}\right)}$ (D) $\frac{1}{e-1}$

Q185. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

The position of a moving car at time t is
given by $f(t) = at^2 + bt + c$, $t > 0$, where a, b and c are real numbers greater than 1. Then the average speed of the car over the time interval $[t_1, t_2]$ is attained at the point :

(A) $\frac{(t_1+t_2)}{2}$ (B) $\frac{(t_2-t_1)}{2}$

(C) $2a(t_1 + t_2) + b$ (D) $a(t_2 - t_1) + b$

Q186. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

Which of the following points lies on the tangent to the curve

$x^4e^y + 2\sqrt{y+1} = 3$ at the point (1, 0)?

(A) (2, 2) (B) (−2, 4) (C) (2, 6) (D) (−2, 6)

Q187. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

The equation of the normal to the curve
$y = (1+x)^{2y} + \cos^2(\sin^{-1}x)$ at x = 0 is :

(A) y = 4x + 2 (B) x + 4y = 8 (C) y + 4x = 2 (D) 2y + x = 4

Q188. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

If the tangent to the curve y = x + sin y at a point
(a, b) is parallel to the line joining $\left(0, \frac{3}{2}\right)$ and $\left(\frac{1}{2}, 2\right)$, then :

(A) b = a (B) |b - a| = 1 (C) $b = \frac{\pi}{2} + a$ (D) |a + b| = 1

Q189. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

Let P(h, k) be a point on the curve
$y = x^2 + 7x + 2$, nearest to the line, y = 3x − 3.
Then the equation of the normal to the curve at P is :

(A) x − 3y − 11 = 0 (B) x − 3y + 22 = 0 (C) x + 3y − 62 = 0 (D) x + 3y + 26 = 0

Q190. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

The length of the perpendicular from the origin, on the normal to the curve,
$x^2 + 2xy - 3y^2 = 0$ at the point (2,2) is

(A) $\sqrt{2}$ (B) $4\sqrt{2}$ (C) 2 (D) $2\sqrt{2}$

Q191. JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

If c is a point at which Rolle's theorem holds for the function,
$f(x) = \log_e\left(\frac{x^2+\alpha}{7x}\right)$ in the interval [3, 4], where $a \in R$, then f"(c) is equal to

(A) $\frac{1}{12}$ (B) $\frac{\sqrt{3}}{7}$ (C) $-\frac{1}{12}$ (D) $-\frac{1}{24}$

Q192. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

The value of c in the Lagrange's mean value theorem for the function
$f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is:

(A) $\frac{2}{3}$ (B) $\frac{\sqrt{7}-2}{3}$ (C) $\frac{4-\sqrt{5}}{3}$ (D) $\frac{4-\sqrt{7}}{3}$

Q193. JEE Main 2020 (Online) 7th January Morning Slot · MCQ (Single Correct Answer)

Let the function, f:[-7, 0]→R be continuous on [-7,0] and differentiable on (-7, 0). If f(-7) = - 3 and f'(x) $\leq$ 2, for all x $\in$ (-7,0), then for all such functions f, f(-1) + f(0) lies in the interval:

(A) $[-6, 20]$ (B) $(-\infty, 20]$

(C) $[-3, 11]$ (D) $(-\infty, 11]$

Q194. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

If the tangent to the curve $y = \frac{x}{x^2-3}$, $x \in \rho$, $\left(x \neq \pm\sqrt{3}\right)$, at a point $(\alpha, \beta) \neq (0, 0)$ on it is parallel to the line 2x + 6y − 11 = 0, then :

(A) $|\ 6\alpha + 2\beta\ | = 9$ (B) $|\ 2\alpha + 6\beta\ | = 11$

(C) $|\ 2\alpha + 6\beta\ | = 19$ (D) $|\ 6\alpha + 2\beta\ | = 19$

Q195. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Let S be the set of all values of x for which the tangent to the curve y = f(x) = $x^3 - x^2 - 2x$ at (x, y) is parallel to the line segment joining the points (1, f(1)) and (−1, f(−1)), then S is equal to :

(A) $\left\{\frac{1}{3}, -1\right\}$ (B) $\left\{-\frac{1}{3}, 1\right\}$

(C) $\left\{-\frac{1}{3}, -1\right\}$ (D) $\left\{\frac{1}{3}, 1\right\}$

Q196. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the tangent to the curve, $y = x^3 + ax - b$ at the point (1, −5) is perpendicular to the line, −x + y + 4 = 0, then which one of the following points lies on the curve ?

(A) (2, −2) (B) (2, −1) (C) (−2, 2) (D) (−2, 1)

Q197. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

The tangent to the curve $y = x^2 - 5x + 5$, parallel to the line 2y = 4x + 1, also passes through the point :

(A) $\left\{\frac{1}{4}, \frac{7}{2}\right\}$ (B) $\left(-\frac{1}{8}, 7\right)$

(C) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (D) $\left(\frac{1}{8}, -7\right)$

Q198. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

The tangent to the curve, $y = xe^{x^2}$ passing through the point (1, e) also passes through the point

(A) $\left(\frac{4}{3}, 2e\right)$ (B) (3, 6e) (C) (2, 3e) (D) $\left(\frac{5}{3}, 2e\right)$

Q199. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

If the curves $y^2 = 6x$, $9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is :

(A) $\frac{9}{2}$ (B) 6 (C) $\frac{7}{2}$ (D) 4

Q200. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

A tangent to the curve, y = f(x) at P(x, y) meets x-axis at A and y-axis at B. If AP : BP = 1 : 3 and f(1) = 1, then the curve also passes through the point :

(A) $\left(\frac{1}{3}, 24\right)$ (B) $\left(\frac{1}{2}, 4\right)$ (C) $\left(2, \frac{1}{8}\right)$ (D) $\left(3, \frac{1}{28}\right)$

Q201. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The tangent at the point $(2, -2)$ to the curve, $x^2y^2 - 2x = 4(1 - y)$ **does not** pass through the point :

(A) $\left(4, \frac{1}{3}\right)$ (B) (8, 5) (C) $(-4, -9)$ (D) $(-2, -7)$

Q202. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The normal to the curve y(x − 2)(x − 3) = x + 6 at the point where the curve intersects the y-axis passes through the point :

(A) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, -\frac{1}{3}\right)$
(C) $\left(\frac{1}{2}, \frac{1}{3}\right)$ (D) $\left(-\frac{1}{2}, -\frac{1}{3}\right)$

Q203. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

Let C be a curve given by $y(x) = 1 + \sqrt{4x - 3}, x > \frac{3}{4}$. If P is a point on C, such that the tangent at P has slope $\frac{2}{3}$, then a point through which the normal at P passes, is :

(A) (2, 3) (B) (4, −3) (C) (1, 7) (D) (3, − 4),

Q204. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the tangent at a point P, with parameter t, on the curve $x = 4t^2 + 3$, $y = 8t^3 - 1$, $t \in \mathbf{R}$, meets the curve again at a point Q, then the coordinates of Q are :

(A) $(t^2 + 3, -t^3 - 1)$ (B) $(4t^2 + 3, -8t^3 - 1)$
(C) $(t^2 + 3, t^3 - 1)$ (D) $(16t^2 + 3, -64t^3 - 1)$

Q205. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

Consider :

f $(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right), x \in \left(0, \frac{\pi}{2}\right)$.

A normal to $y =$ f(x) at $x = \frac{\pi}{6}$ also passes through the point:

(A) $\left(\frac{\pi}{6}, 0\right)$ (B) $\left(\frac{\pi}{4}, 0\right)$

(C) $(0, 0)$ (D) $\left(0, \frac{2\pi}{3}\right)$

Q206. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The normal to the curve, $x^2 + 2xy - 3y^2 = 0$, at $(1, 1)$

(A) meets the curve again in the third quadrant.

(B) meets the curve again in the fourth quadrant.

(C) does not meet the curve again.

(D) meets the curve again in the second quadrant.

Q207. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1), g(0) = 0$ and $f(1) = 6$, then for some $c \in]0, 1[$

(A) $f'(c) = g'(c)$ (B) $f'(c) = 2g'(c)$

(C) $2f'(c) = g'(c)$ (D) $2f'(c) = 3g'(c)$

Q208. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

The intercepts on x-axis made by tangents to the curve, $y = \int_0^x |t| dt, x \in R$, which are parallel to the line $y = 2x$, are equal to :

(A) ± 1 (B) ± 2 (C) ± 3 (D) ± 4

Q209. AIEEE 2011 · MCQ (Single Correct Answer)

The shortest distance between line $y - x = 1$ and curve $x = y^2$ is

(A) $\frac{3\sqrt{2}}{8}$ (B) $\frac{8}{3\sqrt{2}}$ (C) $\frac{4}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{4}$

Q210. AIEEE 2010 · MCQ (Single Correct Answer)

The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to the x-axis, is

(A) $y = 1$ (B) $y = 2$ (C) $y = 3$ (D) $y = 0$

Q211. AIEEE 2007 · MCQ (Single Correct Answer)

A value of c for which conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is

(A) $\log_3 e$ (B) $\log_e 3$ (C) $2\log_3 e$ (D) $\frac{1}{2}\log_3 e$

Q212. AIEEE 2006 · MCQ (Single Correct Answer)

Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points $(2, 0)$ and $(3, 0)$ is

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Q213. AIEEE 2005 · MCQ (Single Correct Answer)

The normal to the curve
$x = a(\cos\theta + \theta\sin\theta), y = a(\sin\theta - \theta\cos\theta)$ at any point
θ' is such that

(A) it passes through the origin

(B) it makes an angle $\frac{\pi}{2} + \theta$ with the x-axis

(C) it passes through $\left(a\frac{\pi}{2}, -a\right)$

(D) it is at a constant distance from the origin

Q214. AIEEE 2005 · MCQ (Single Correct Answer)

If the equation $a_n x^n + a_{n-1}x^{n-1} + \ldots\ldots\ldots + a_1 x = 0$
$a_1 \neq 0, n \geq 2$, has a positive root $x = \alpha$, then the equation
$na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \ldots\ldots\ldots + a_1 = 0$ has a positive root, which is

(A) greater than α

(B) smaller than α

(C) greater than or equal to smaller than α

(D) equal to smaller than α

Q215. AIEEE 2005 · MCQ (Single Correct Answer)

Let f be differentiable for all x. If f(1) = -2 and f'(x) $\geq$ 2 for
x $\in [1, 6]$, then

(A) f(6) $\geq$ 8 (B) f(6) < 8 (C) f(6) < 5 (D) f(6) = 5

Q216. AIEEE 2004 · MCQ (Single Correct Answer)

A function $y = f(x)$ has a second order derivative $f''(x) = 6(x-1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function is :

(A) $(x+1)^2$ (B) $(x-1)^3$ (C) $(x+1)^3$ (D) $(x-1)^2$

Q217. AIEEE 2004 · MCQ (Single Correct Answer)

The normal to the curve $x = a(1 + \cos\theta)$, $y = a\sin\theta$ at 'θ' always passes through the fixed point

(A) (a, a) (B) $(0, a)$ (C) $(0, 0)$ (D) $(a, 0)$

Q218. AIEEE 2004 · MCQ (Single Correct Answer)

If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval

(A) $(1, 3)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) $(0, 1)$

Q219. AIEEE 2002 · MCQ (Single Correct Answer)

If $2a + 3b + 6c = 0$, $(a, b, c \in R)$ then the quadratic equation $ax^2 + bx + c = 0$ has

(A) at least one root in $[0, 1]$ (B) at least one root in $[2, 3]$

(C) at least one root in $[4, 5]$ (D) none of these

Q220. JEE Main 2023 (Online) 10th April Evening Shift · Numerical

Let the quadratic curve passing through the point $(-1, 0)$ and touching the line $y = x$ at $(1, 1)$ be $y = f(x)$. Then the x-intercept of the normal to the curve at the point $(\alpha, \alpha + 1)$ in the first quadrant is _____.

Q221. JEE Main 2023 (Online) 6th April Evening Shift · Numerical

Let a curve $y = f(x), x \in (0, \infty)$ pass through the points $P\left(1, \frac{3}{2}\right)$ and $Q\left(a, \frac{1}{2}\right)$. If the tangent at any point $R(b, f(b))$ to the given curve cuts the y-axis at the point $S(0, c)$ such that $bc = 3$, then $(PQ)^2$ is equal to _____.

Q222. JEE Main 2023 (Online) 29th January Evening Shift · Numerical

If the equation of the normal to the curve $y = \frac{x-a}{(x+b)(x-2)}$ at the point (1, -3) is $x - 4y = 13$, then the value of $a + b$ is equal to _____.

Q223. JEE Main 2022 (Online) 29th July Evening Shift · Numerical

If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a, b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point (2, -1), then $|2a + 9b|$ is equal to _____.

Q224. JEE Main 2022 (Online) 27th July Morning Shift · Numerical

Let M and N be the number of points on the curve $y^5 - 9xy + 2x = 0$, where the tangents to the curve are parallel to x-axis and y-axis, respectively. Then the value of $M + N$ equals _____.

Q225. JEE Main 2022 (Online) 28th June Morning Shift · **Numerical**

Let l be a line which is normal to the curve $y = 2x^2 + x + 2$ at a point P on the curve. If the point Q(6, 4) lies on the line l and O is origin, then the area of the triangle OPQ is equal to _____.

Q226. JEE Main 2021 (Online) 26th February Evening Shift · **Numerical**

Let the normals at all the points on a given curve pass through a fixed point (a, b). If the curve passes through (3, -3) and (4, $-2\sqrt{2}$), and given that $a - 2\sqrt{2}\, b = 3$,
then $(a^2 + b^2 + ab)$ is equal to _____.

Q227. JEE Main 2021 (Online) 25th February Evening Shift · **Numerical**

If the curves $x = y^4$ and $xy = k$ cut at right angles, then $(4k)^6$ is equal to _____.

Q228. JEE Main 2020 (Online) 5th September Evening Slot · **Numerical**

If the lines $x + y = a$ and $x - y = b$ touch the
curve $y = x^2 - 3x + 2$ at the points where the curve intersects the x-axis, then $\frac{a}{b}$ is equal to _____.

Q229. JEE Main 2020 (Online) 8th January Morning Slot · **Numerical**

Let the normal at a point P on the curve
$y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0, \frac{3}{2}\right)$.
If m is the slope of the tangent at P to the curve, then |m| is equal to

Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **D**	3. **C**	4. **B**	5. **A**
6. **B**	7. **D**	8. **B**	9. **C**	10. **B**
11. **A**	12. **D**	13. **B**	14. **A**	15. **B**
16. **B**	17. **A**	18. **B**	19. **D**	20. **C**
21. **A**	22. **D**	23. **A**	24. **B**	25. **D**
26. **D**	27. **D**	28. **D**	29. **B**	30. **A**
31. **C**	32. **B**	33. **A**	34. **C**	35. **A**
36. **A**	37. **B**	38. **A**	39. **C**	40. **A**
41. **C**	42. **B**	43. **C**	44. **C**	45. **A**
46. **A**	47. **D**	48. **C**	49. **A**	50. **D**
51. **A**	52. **C**	53. **A**	54. **C**	55. **B**
56. **A**	57. **A**	58. **D**	59. **B**	60. **B**
61. **B**	62. **A**	63. **B**	64. **C**	65. **D**
66. **A**	67. **C**	68. **B**	69. **A**	70. **C**
71. **C**	72. **D**	73. **C**	74. **D**	75. **C**
76. **C**	77. **D**	78. **A**	79. **A**	80. **A**
81. **D**	82. **C**	83. **C**	84. **A**	85. **D**
86. **A**	87. **C**	88. **B**	89. **D**	90. **A**
91. **B**	92. **A**	93. **C**	94. **A**	95. **A**
96. **B**	97. **B**	98. **D**	99. **D**	100. **B**
101. **C**	102. **B**	103. **D**	104. **B**	105. **C**
106. **A**	107. **C**	108. **A**	109. **D**	110. **A**
111. **B**	112. **A**	113. **D**	114. **A**	115. **B**
116. **B**	117. **C**	118. **A**	119. **A**	120. **A**
121. **D**	122. **C**	123. **B**	124. **C**	125. **C**
126. **C**	127. **D**	128. **A**	129. **A**	130. **B**
131. **D**	132. **C**	133. **A**	134. **C**	135. **A**
136. **B**	137. **C**	138. **C**	139. **A**	140. **C**
141. **D**	142. **B**			

Numerical

143. **4**	144. **1**	145. **3**	146. **30**	147. **252**
148. **39**	149. **128**	150. **1600**	151. **5**	152. **2890**
153. **48**	154. **5**	155. **5**	156. **5**	157. **45**
158. **15**	159. **4**	160. **3**	161. **22**	162. **2**
163. **4**	164. **36**	165. **5**	166. **2**	167. **144**
168. **9**	169. **3**			

Out of syllabus

170. **D**	171. **C**	172. **C**	173. **D**	174. **C**
175. **D**	176. **B**	177. **A**	178. **C**	179. **D**
180. **D**	181. **C**	182. **B**	183. **C**	184. **C**
185. **A**	186. **D**	187. **B**	188. **B**	189. **D**
190. **D**	191. **A**	192. **D**	193. **B**	194. **D**
195. **B**	196. **A**	197. **D**	198. **A**	199. **A**
200. **C**	201. **D**	202. **A**	203. **C**	204. **A**
205. **D**	206. **B**	207. **B**	208. **A**	209. **A**
210. **C**	211. **C**	212. **B**	213. **D**	214. **B**
215. **A**	216. **B**	217. **D**	218. **D**	219. **A**
220. **11**	221. **5**	222. **4**	223. **195**	224. **2**
225. **13**	226. **9**	227. **4**	228. **0.50**	229. **4**